



Greedy Algorithm



Lalach buri bala NAHI hai! (Ek fun Hinglish Guide)

1. Greedy Algorithm kya hota hai? 🤔

Bachpan mein humne suna tha "Lalach buri bala hai", par DSA (Data Structures & Algorithms) ki duniya mein **Lalach (Greed) achhi cheez hai!**

Greedy Algorithm ka simple rule: *"Abhi jo samne sabse bada fayda dikh raha hai, usko utha lo! Future ki tension mat lo."* Local best solution ko pakdo, taaki Global best mil jaye.

🏏 Cricket vs ♟️ Chess (Real-life Example)

- **Cricket (Greedy Hai):** Har ball par maximum run (Sixer) maarne ki koshish karte ho. Tum nahi sochte ki agli ball kaisi aayegi. Har ball pe lalach = Greedy Approach!
- **Chess (Greedy NAHI Hai):** Chess mein kabhi-kabhi apna pyada (pawn) marwana padta hai taaki future mein checkmate kar sako. Yahan future dekhna padta hai, isliye ye Greedy nahi hai.

Naukri Job Hunt se connection: Agar tumhe Data Analyst ki ek job mile jo turant bohot achha package de rahi ho, aur tum bina future growth soche usko le lo, toh wo *Greedy Approach* hai. Lekin agar tum temporarily ek low-paying internship lo taaki 2 saal baad direct Google/Microsoft mein ja sako, wo *Dynamic Programming (Future planning)* hui!

2. Array Max/Min Difference Problem



Agar tumhe ek Array diya jaye aur bola jaye ki iske elements ka difference nikalo taaki Sum ek baar **Maximum** aaye aur ek baar **Minimum** aaye. Kaise karoge?

The Key Rule: SORTING!

Greedy ka sabse bada dost hai **Sorting**. Jab bhi Maximum ya Minimum nikalna ho, sabse pehle array ko sort kar do!

▲ **Maximum Sum chahiye?** Sabse door wale elements ko minus karo. (Sabse bada - Sabse chhota).

[2]  [30]

First element (2) and Last element (30) gives max difference (28)!

▼ **Minimum Sum chahiye?** Padosiyon (neighbors) ko minus karo. Consecutive elements ka difference humesha sabse kam hota hai.

[2]  [5]

Neighbors 5 and 2 give minimum difference (3)!

3. The Coin Change Problem (Brass Lights Business)

Maan lo tumhare **Decorative Brass Lights** ke business mein ek customer ne ₹1024 ka bill cash mein pay kiya. Ab tumhe change wapas karna ya manage karna hai minimum notes mein. Tumhare paas har type ke notes unlimited hain!

Lalach wala dimaag (Greedy Algorithm): Tum sabse pehle sabse bada note uthaoge!

1. Pehle ₹500 ka note liya (Bache ₹524)
2. Ek aur ₹500 ka note liya (Bache ₹24)
3. Phir ₹20 ka note liya (Bache ₹4)
4. Phir ₹2 ke do sikke (coins) liye!

Total coins/notes = 5 (Minimum possible!). Tumne yahan sorting ki (bade se chhota) aur greedy ho kar sabse bada amount pehle chuna.

4. Python Code (PyCharm Ready!)

Ye raha Array Max/Min Difference ka simple aur sweet Python code:

```
def max_min_diff(arr):
    # Step 1: Sorting - Greedy ka best friend!
    arr.sort()
    n = len(arr)
    mid = n // 2

    max_diff = 0
    min_diff = 0

    # Do pointers: 'i' start se, 'j' end se
    j = n - 1

    # Loop sirf aadhe array tak chalega
    for i in range(mid):
        # Maximum ke liye: Aakhiri aur pehla element minus karo
        max_diff += abs(arr[i] - arr[j])
        j -= 1

        # Minimum ke liye: Padosi (neighbors) minus karo
        min_diff += abs(arr[2*i] - arr[2*i + 1])

    print(f"Maximum Difference = {max_diff}")
    print(f"Minimum Difference = {min_diff}")

# Chalo test karein!
my_array = [12, 5, 25, 10, 2, 15, 8, 30]
max_min_diff(my_array)
```



Jaise Byomkesh Bakshi ek-ek chote clue (local optimal) ko dhoondh kar final mystery (global optimal) solve karta hai, Greedy Algorithm bhi bilkul waise hi kaam karta hai! 